

BACKGROUND

- Infants can represent relational structures and generalize them to new stimuli within the first year (Anderson et al., 2018; Hespos et al., 2021; Hochmann et al., 2016)
- However, comprehension and productive use of relational words (e.g., same, different) emerge around ages 2.5-3 (Christie & Gentner, 2014)
- By 15-19 months, infants can map words to abstract or non-visible referents (Luchkina & Waxman, 2025)
- If infants can represent and map abstract relations early, **why is learning of relational words so delayed?**
- Perhaps, natural language provides few opportunities to infer relational word meanings due to polysemy and referential ambiguity (e.g., “same dress,” “same day”)
- CHILDES (MacWhinney, 2000) corpus analysis shows that caregivers rarely label abstract relations, unlike objects, which are labeled with concrete nouns
- E.g., the words “same” and “different” were mentioned ~0.72 times, while “car” and “ball” were mentioned ~3 times per 1000 words when infants were 15-19 months
- Can infants learn relational terms when provided with clearer, more informative input?**

METHOD

- Participants:** 103 15-to-19-month-olds ($M=16.17$ mo; 49 girls) recruited and tested via Children Helping Science
- Procedure:** Contrastive cross-situational word learning with explicit labeling and referential cues; infants were randomly assigned to one of 3 training conditions:
 - Control ($N=36$): No familiarization; test only
 - Simple Familiarization ($N=30$): Sixteen alternating same and different image *pairs* labeled “same” or “different” (Figure 1a)
 - Complex Familiarization ($N=37$): Similar to Simple Familiarization, but with varying set sizes to mimic naturalistic ambiguity (Figure 1b)
- Test:** New pairs of animal silhouettes shown side-by-side accompanied by a verbal prompt (Figure 1c)
- Measure:** % of looking time to the target pair 367-2000 ms after the onset of the 2nd mention of the target word

METHOD (continued)



Figure 1a. Simple Familiarization trials. Image sets varied only along the relevant dimension. On each trial, the actor alternated between looking to the images and to the camera to emulate eye contact and produce referential cues.



Figure 1b. Complex Familiarization trials. The size of the image set also varied across trials.

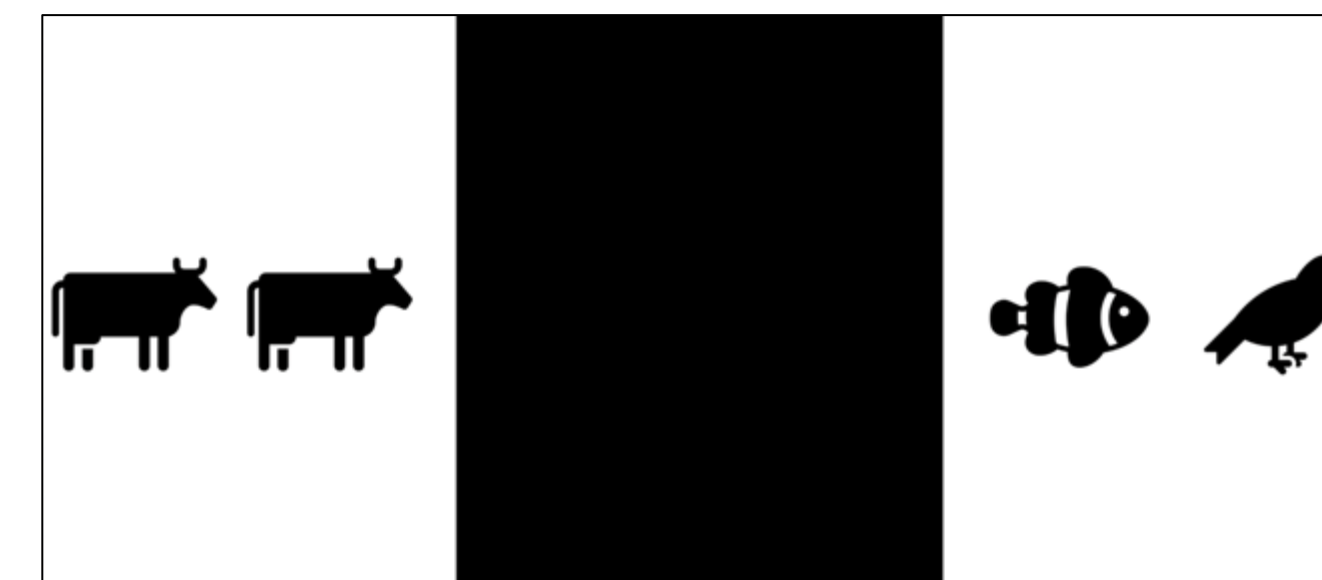


Figure 1c. A test trial, accompanied by a verbal prompt, “Look! They are the same/different, they are the same/different!” The side of the target pair was counterbalanced across trials.

RESULTS

Mean percent of looking to the target pair, averaged across 367-2000 ms after the second mention of the target word

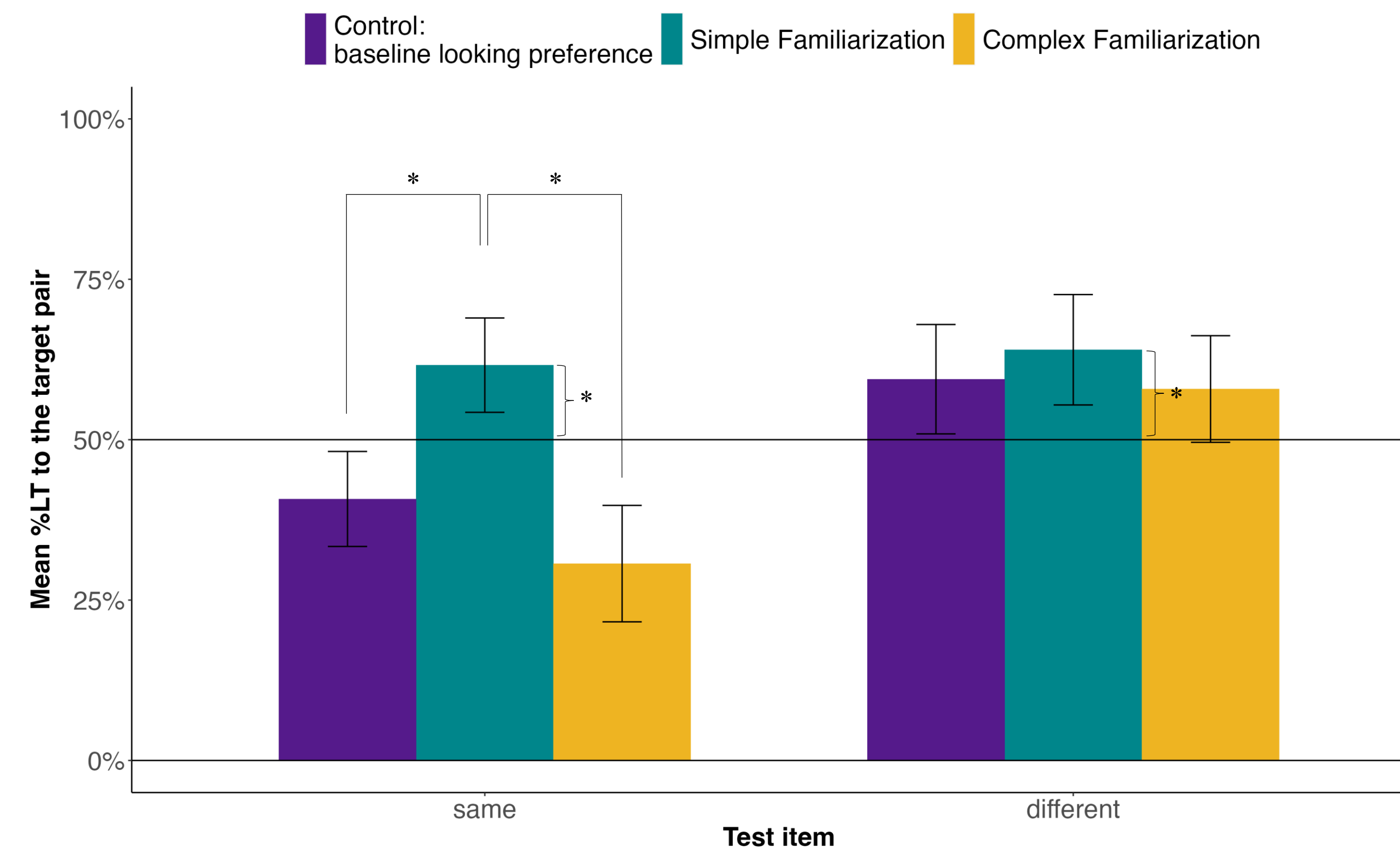


Figure 2. Infants' mean looking preference for the target pair, averaged across the window of analysis.

PREDICTIONS AND ANALYSES

- Predictions:** Infants in the Control and Complex Familiarization conditions would prefer the “different” image sets, **regardless of the label**, since these displays contain more information
- Infants in the Simple Familiarization condition would look to the image set matching the word
- Analyses:** T-tests comparing the proportion of looking to the target to the chance level and across conditions
- Results:** (1) Only infants in the Simple Familiarization condition had a preference for the target on both *same* and *different* trials
- (2) Significant difference between infants' looking to the target in the Simple Familiarization condition and the other two conditions

CONCLUSIONS

- Infants can, in principle, infer and generalize the meanings of relational terms from minimal linguistic input and structured visual input
- The foundations for abstract word learning are present before the second year of life
- The delay in learning may indeed stem from the scarcity and ambiguity of word-referent mappings in natural language exposure

NEXT STEPS AND LIMITATIONS

- Next steps: How do infants generalize the meaning? (e.g., across shapes, colors, functions)
- Limitations: It is an open question how infants represented the relation ‘different’ and whether they learned its name

REFERENCES

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